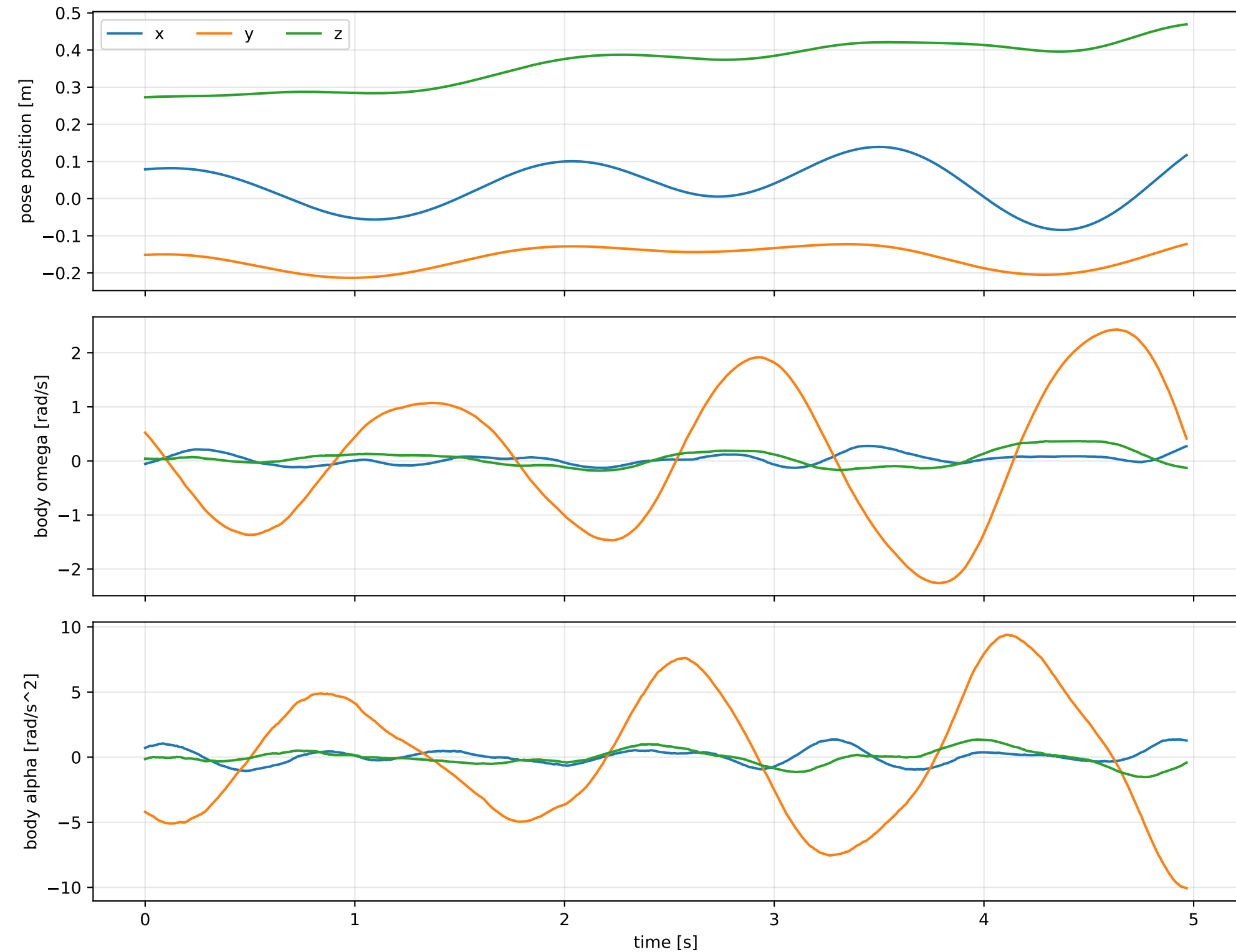
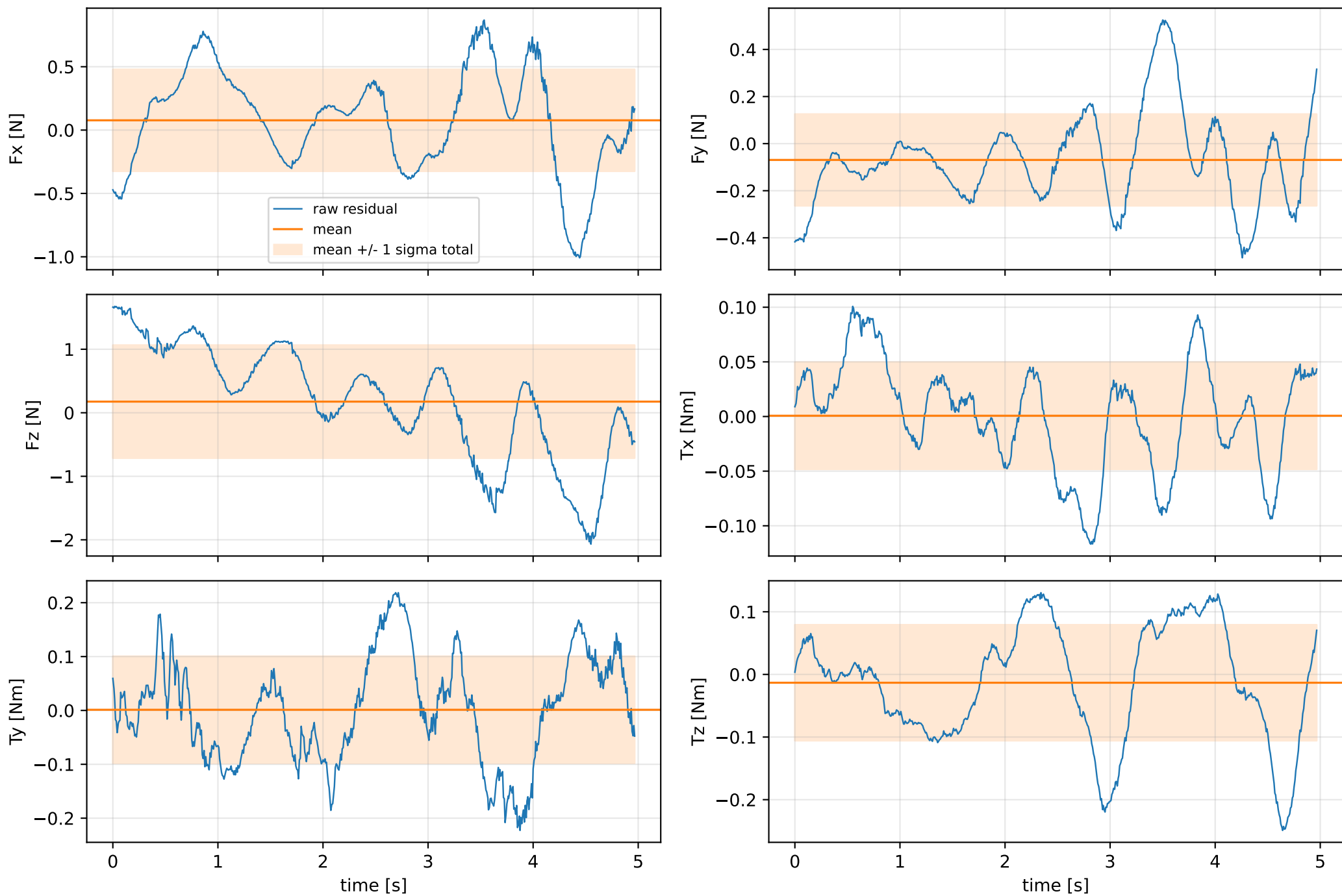


inertia_over_mass_xx_nominal: SG trajectory and derived motion



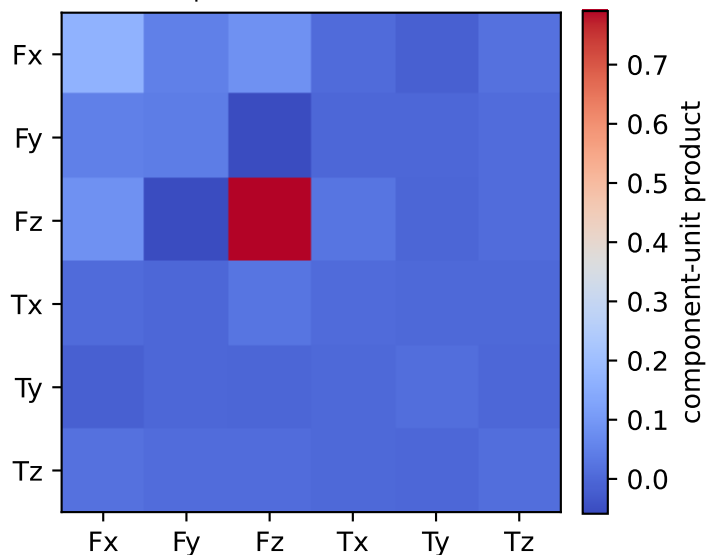
inertia_over_mass_xx_nominal: raw Newton--Euler residual wrench (not trajectory-fitted external wrench)
mass gauge fixed to nominal mass = 2.35159759 kg



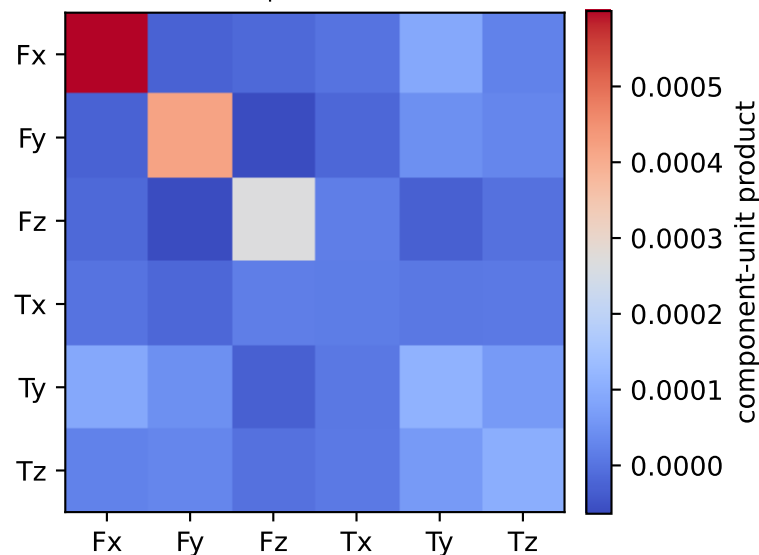
vector RMS about zero: force 1.0147 N, torque 0.14473 Nm; component std about mean: force [0.4018 0.1946 0.8891], torque [0.049 0.0992 0.0925]

inertia_over_mass_xx_nominal: residual-wrench covariance decomposition in nominal-mass gauge
force [N], torque [Nm]; raw excess eigenvalues [0.001 0.006 0.008 0.016 0.174 0.805]

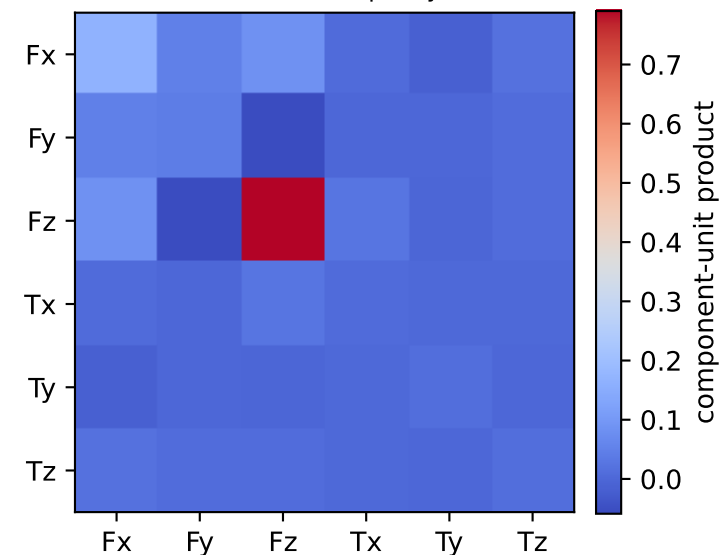
empirical total covariance



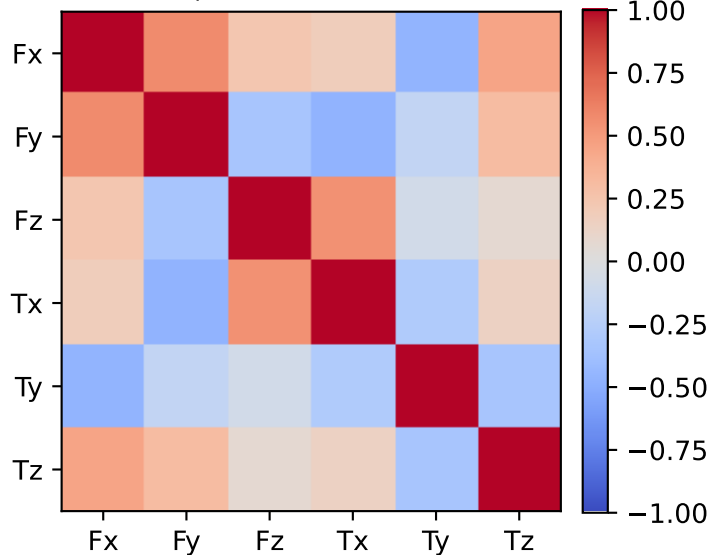
mean full-SG prediction covariance



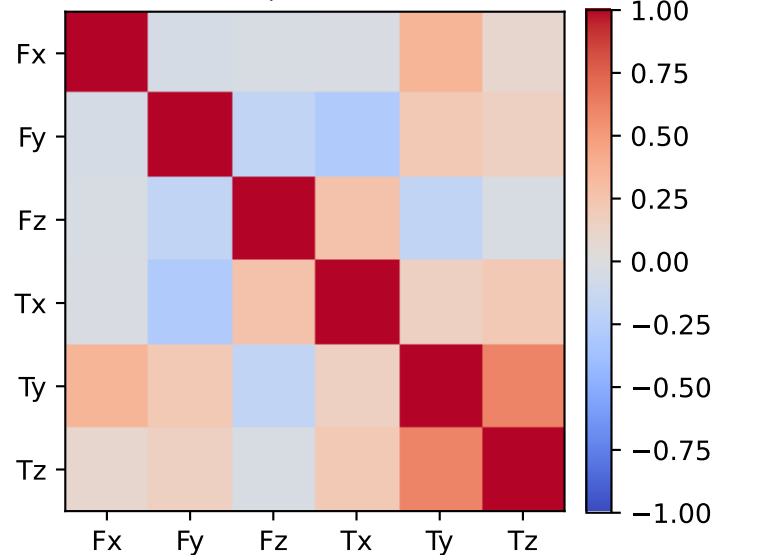
PSD excess model discrepancy covariance



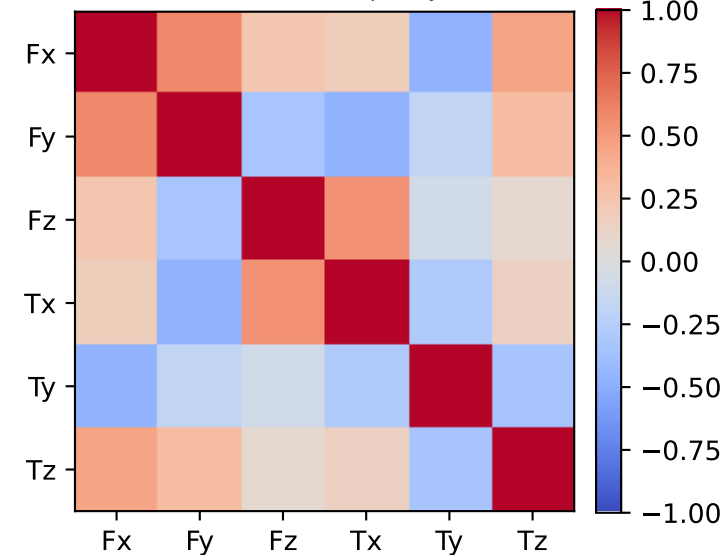
empirical total correlation



mean full-SG prediction correlation

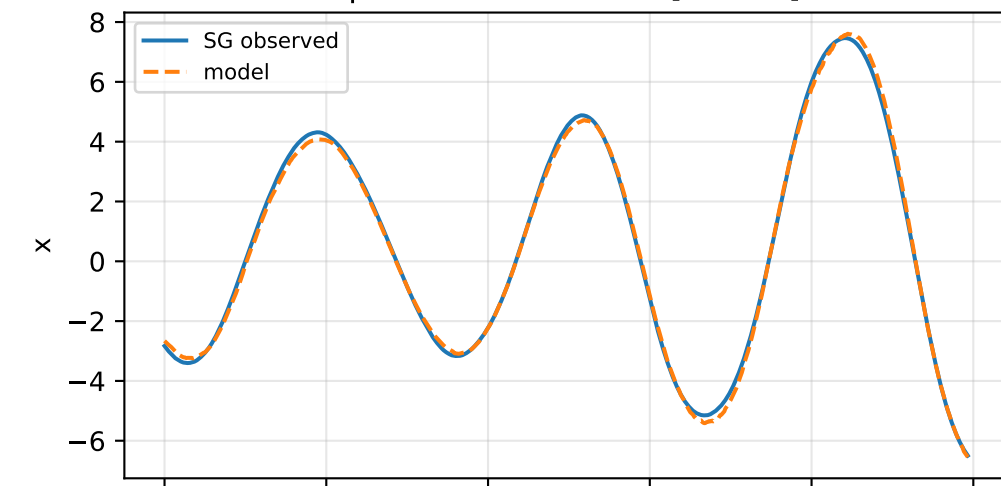


PSD excess model discrepancy correlation

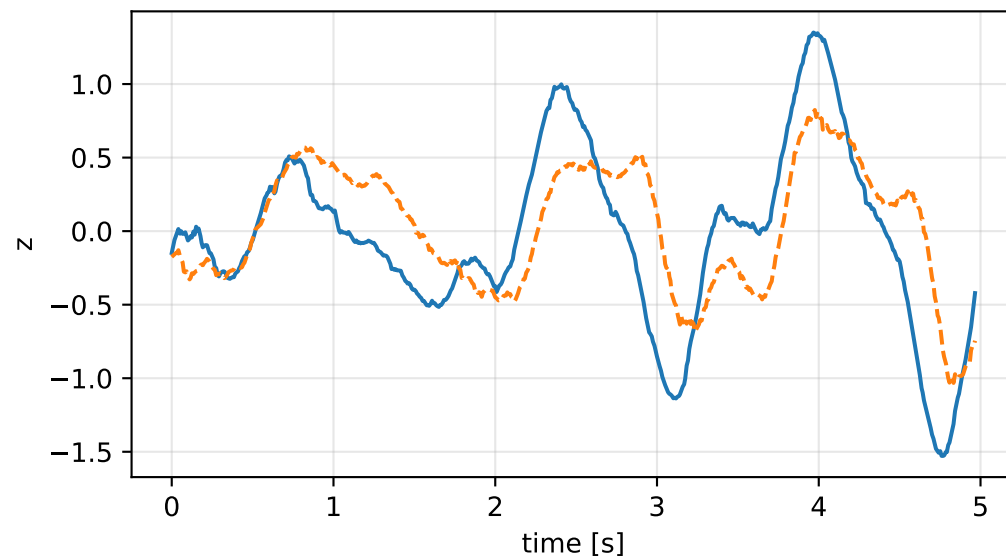
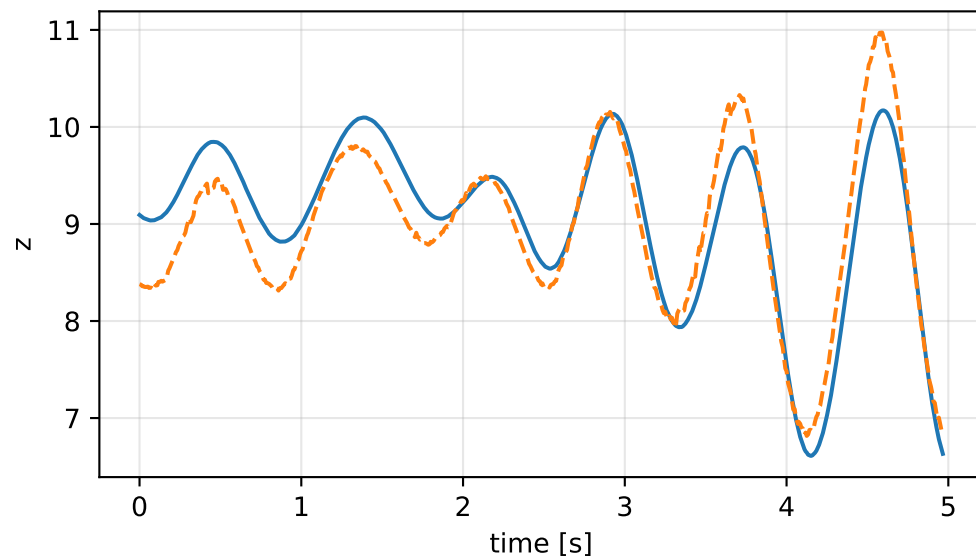
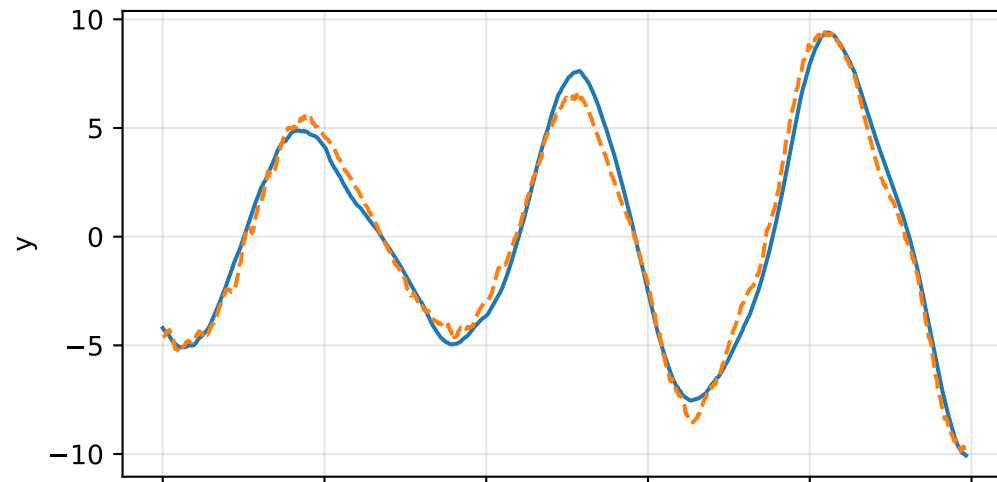
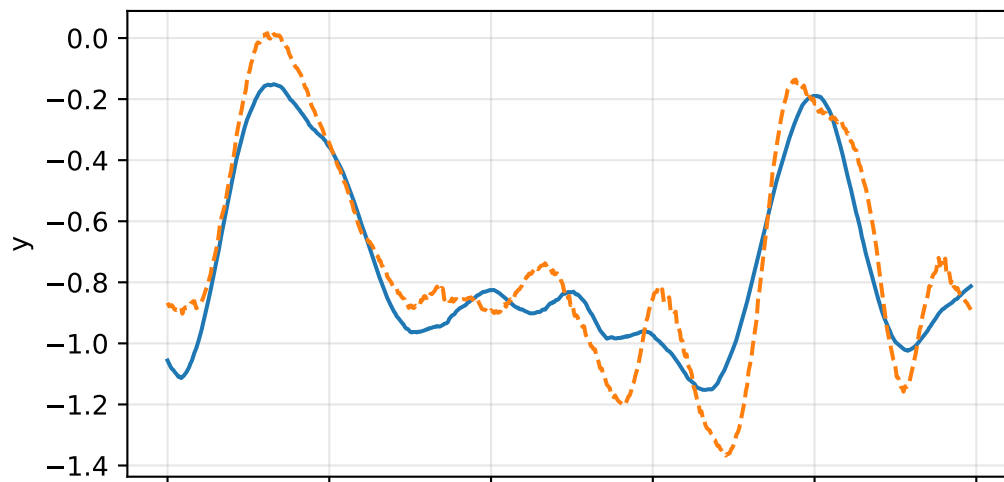
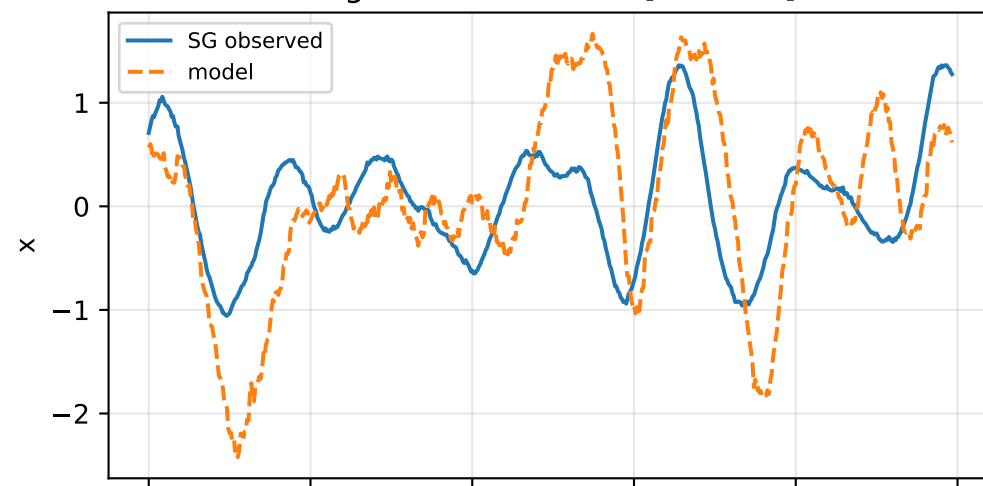


inertia_over_mass_xx_nominal: acceleration residual objective

specific acceleration [m/s^2]

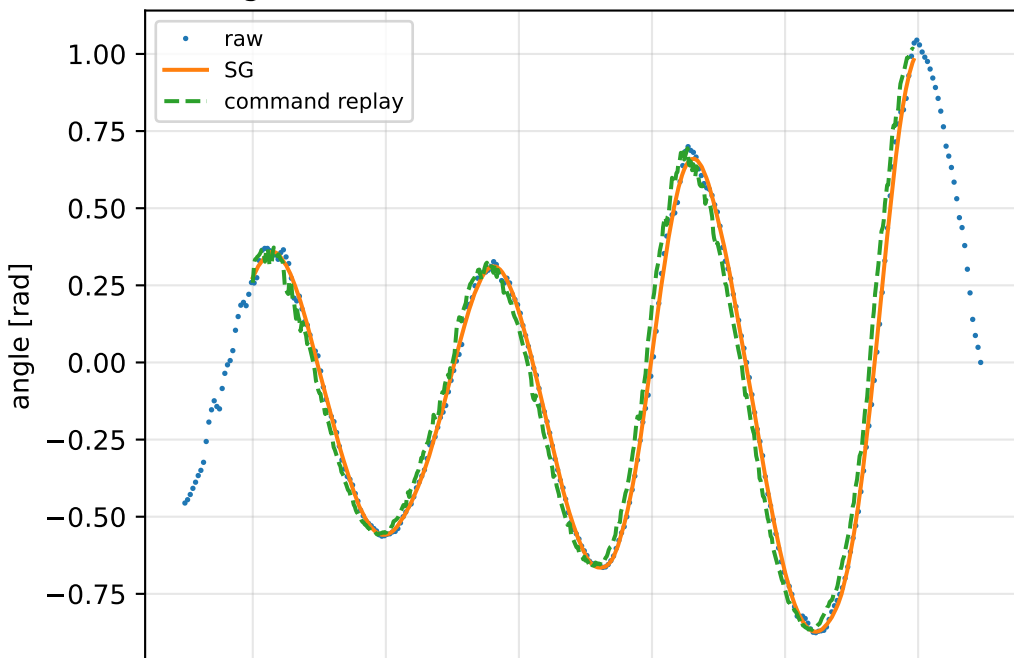


angular acceleration [rad/s^2]

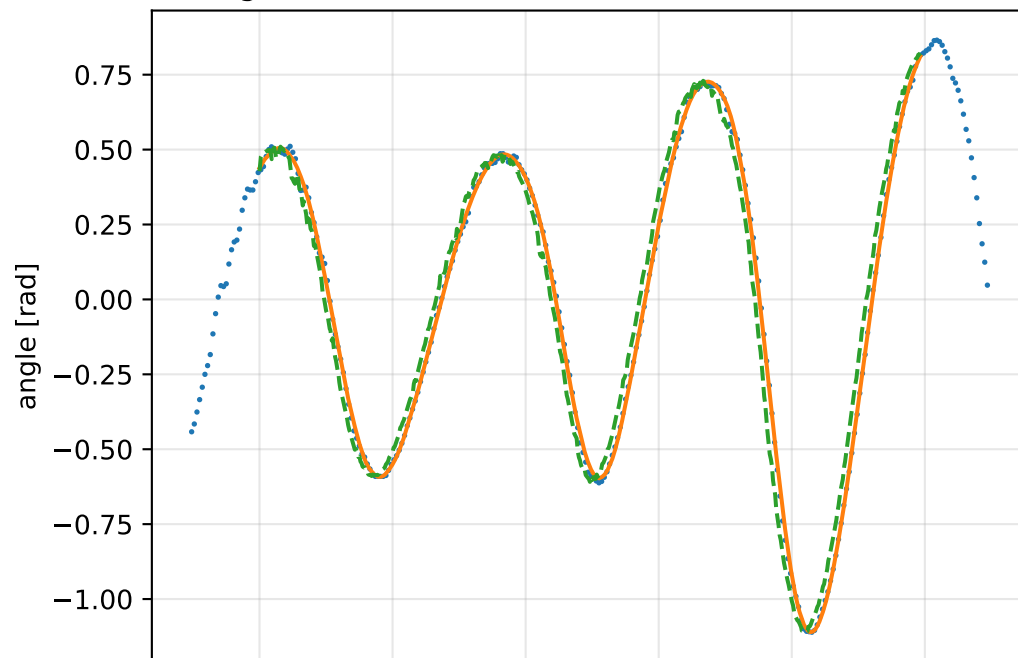


inertia_over_mass_xx_nominal: gimbal measurement audit (objective=measured_sg)

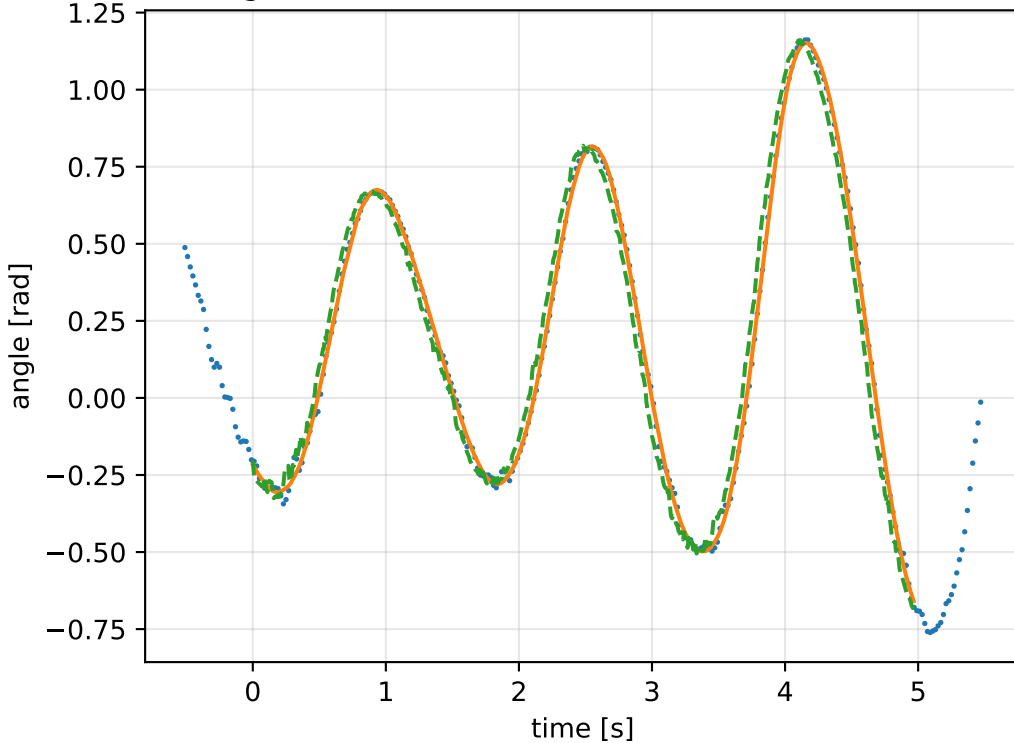
gimbal 1: RMSE=0.0759, max=0.2095 rad



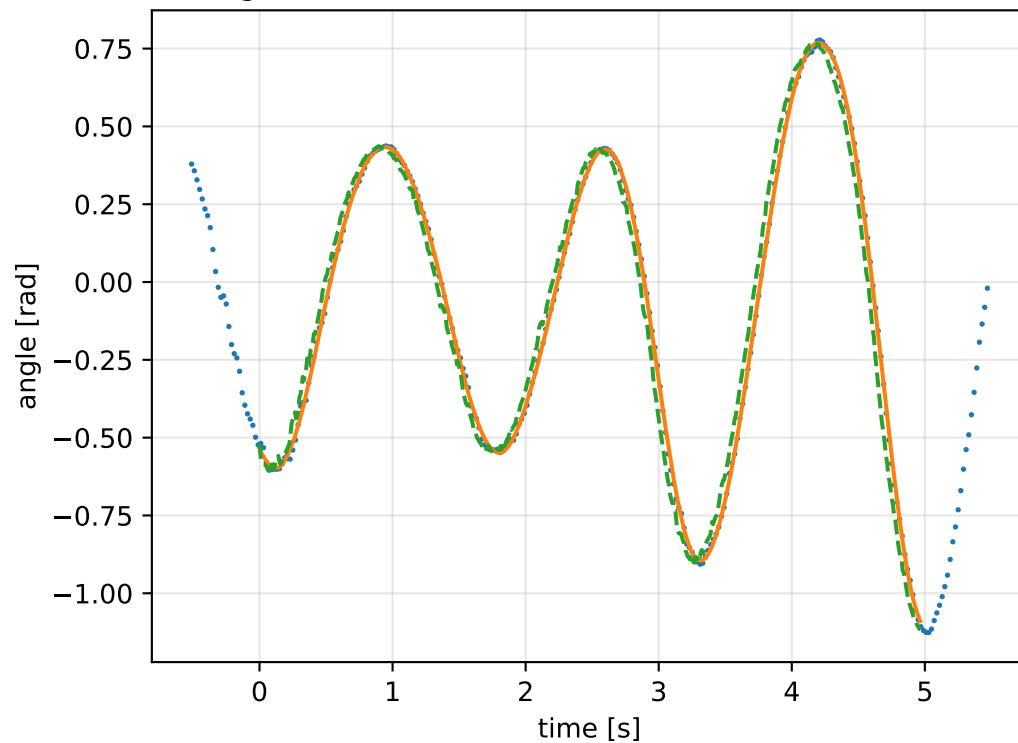
gimbal 2: RMSE=0.08099, max=0.2013 rad



gimbal 3: RMSE=0.07393, max=0.172 rad

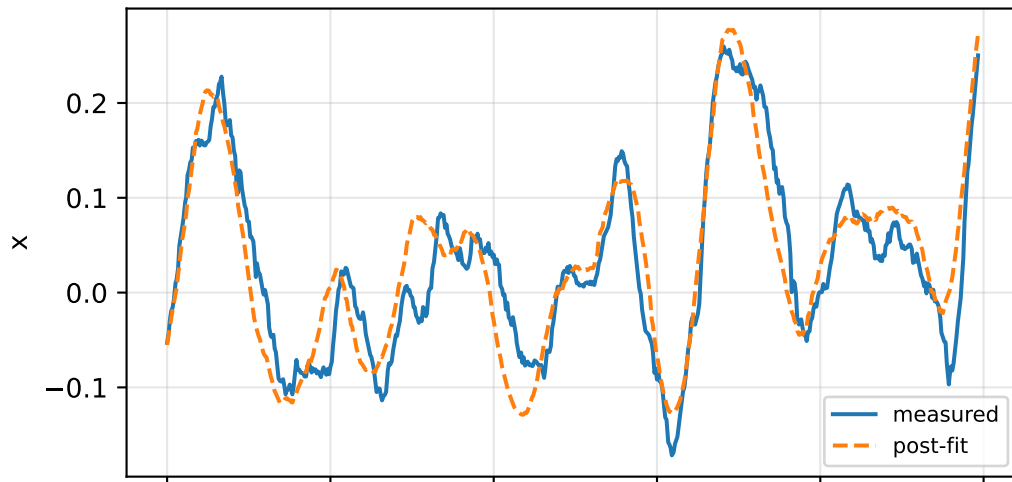


gimbal 4: RMSE=0.07131, max=0.1737 rad

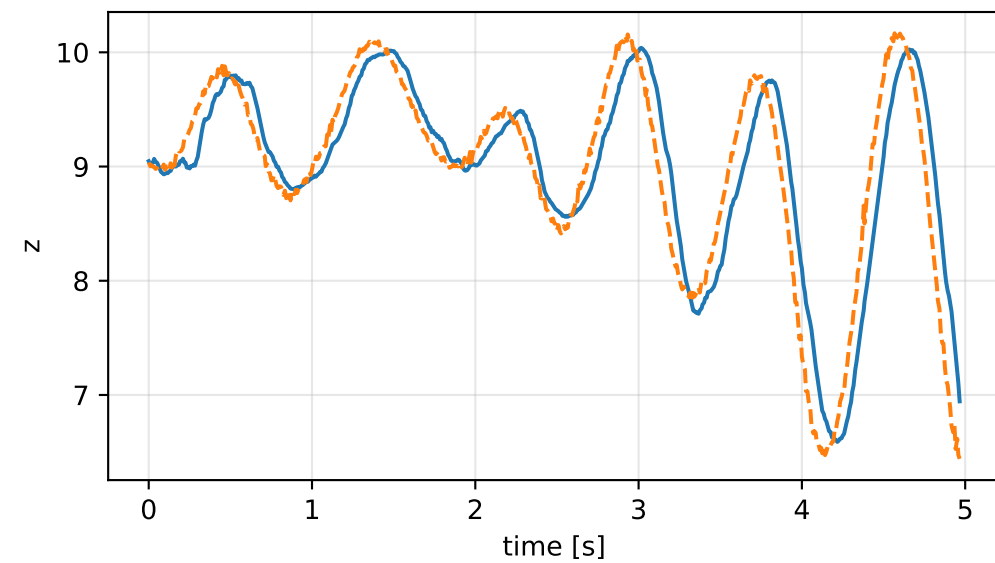
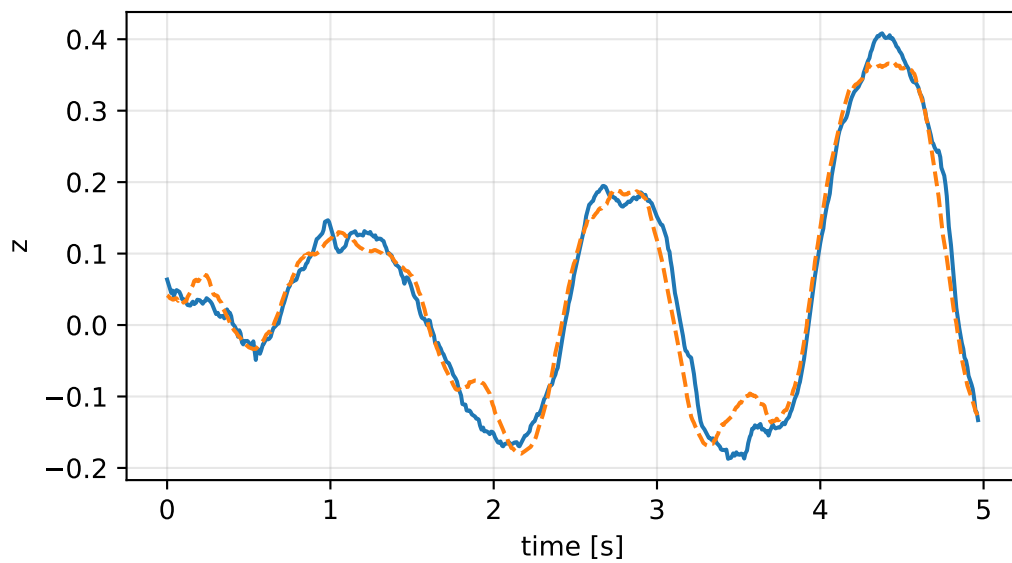
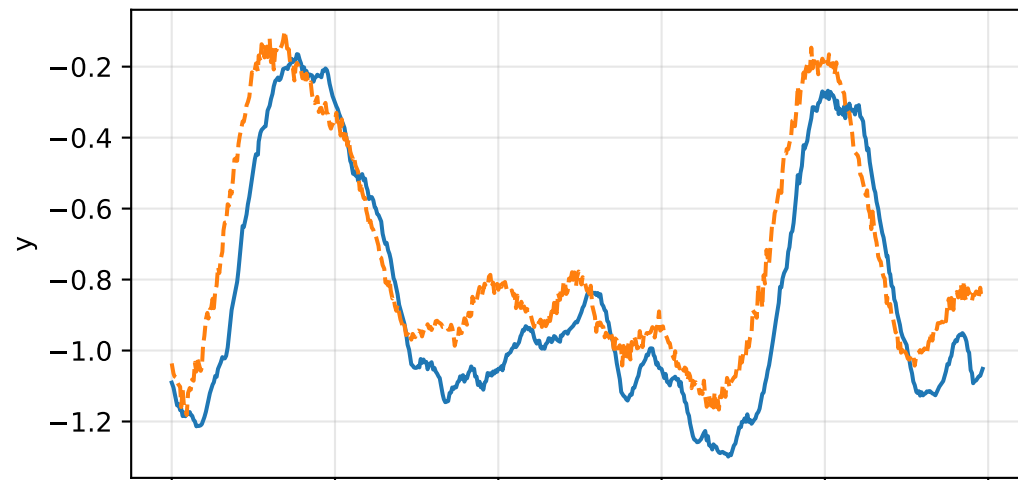
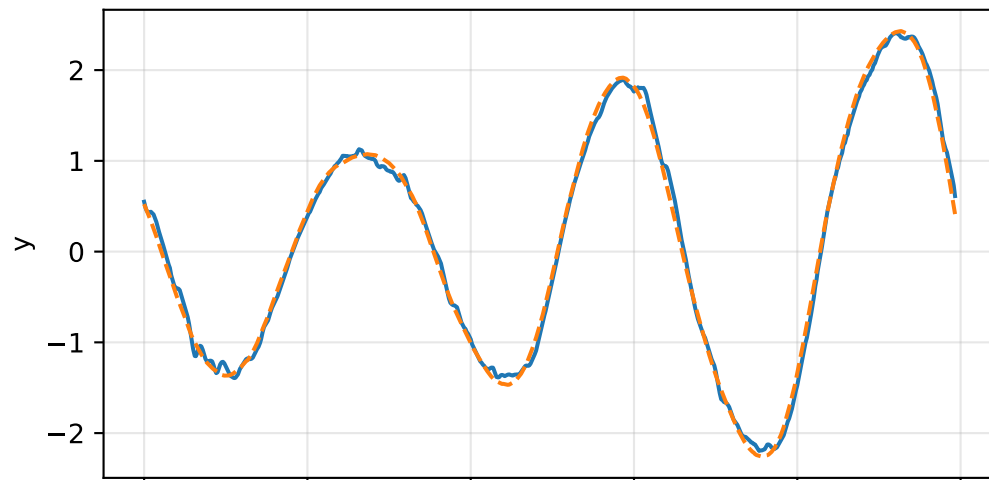
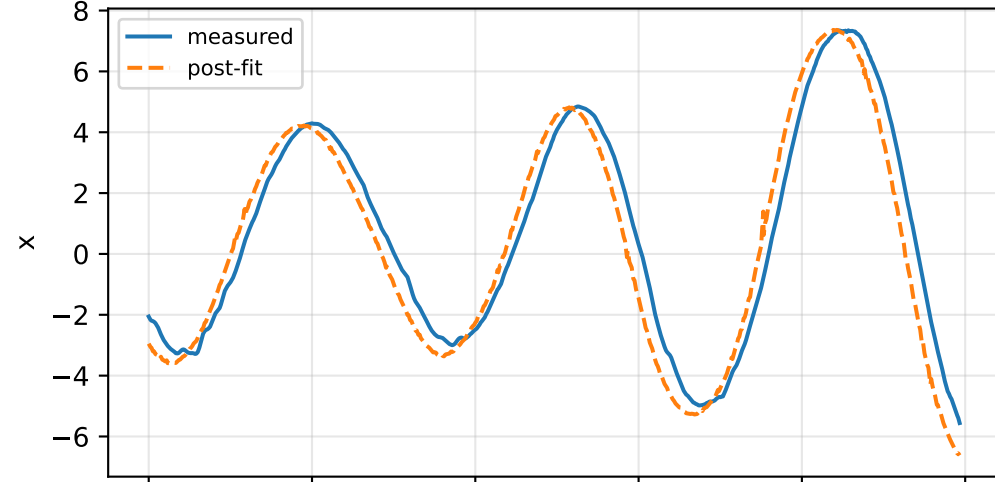


inertia_over_mass_xx_nominal: IMU comparison (diagnostic only)

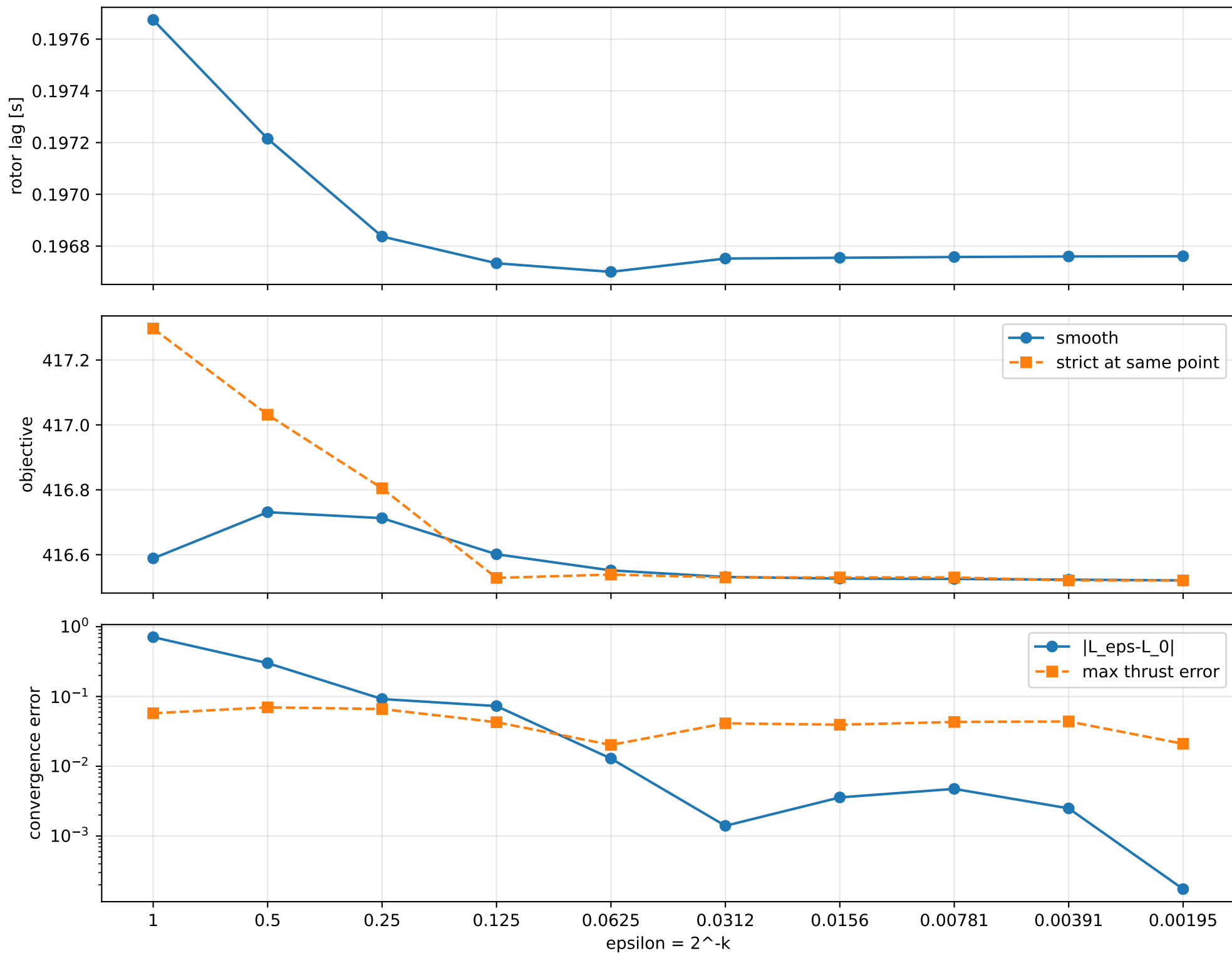
gyro [rad/s]



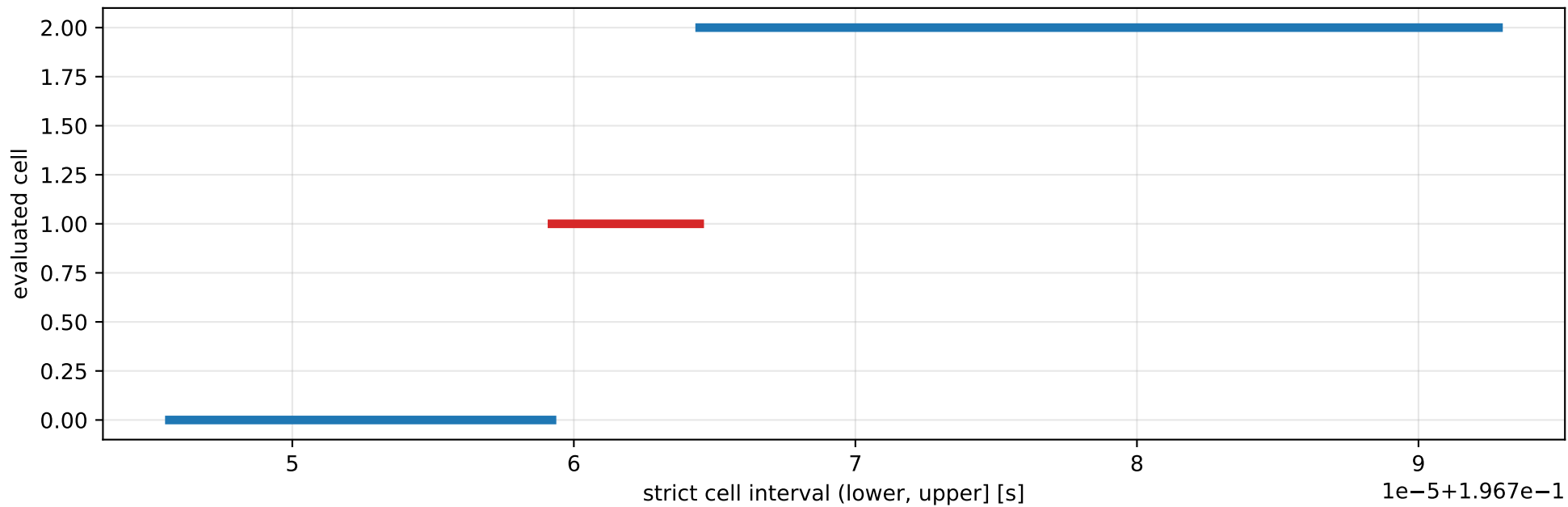
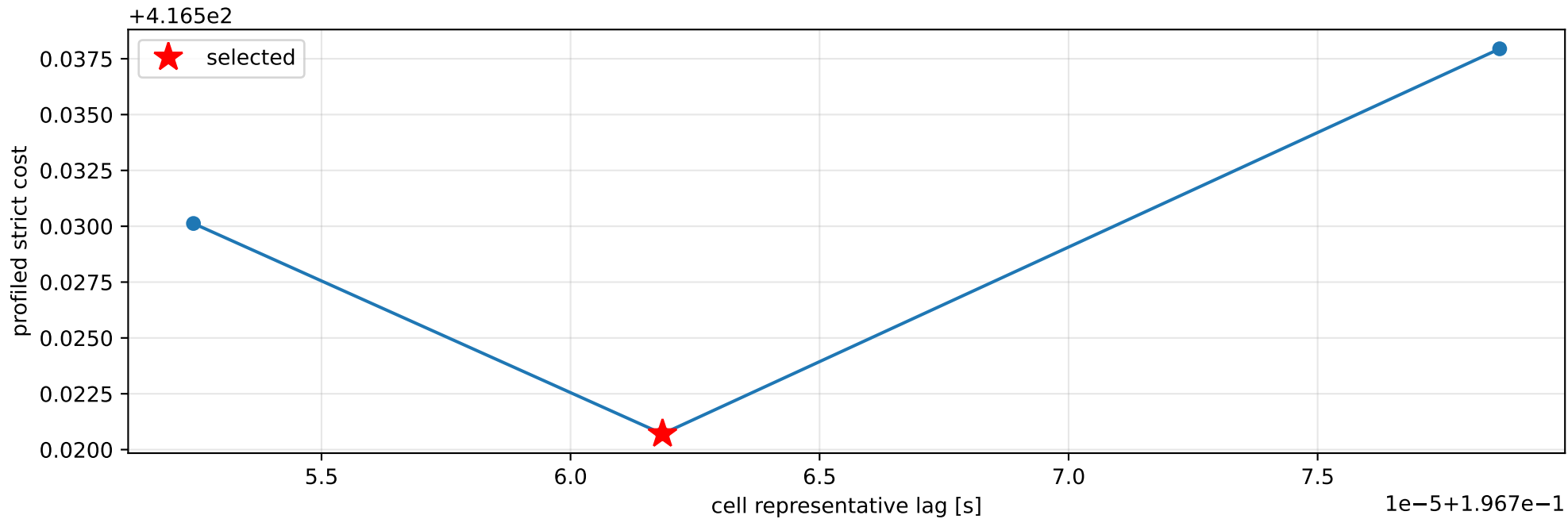
specific force [m/s²]



inertia_over_mass_xx_nominal: smooth-to-strict continuation



inertia_over_mass_xx_nominal: exact strict-ZOH lag cells



selected (0.196759224, 0.196764469] s; final smooth 0.196760854 s; data boundary=False

inertia_over_mass_xx_nominal: parameter estimate

Case: inertia_over_mass_xx_nominal

status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 4.69551624

inertia [kg m²]:

```
[[ 0.12978793 -0.00137075  0.0021837 ]  
 [-0.00137075  0.29142588 -0.0110491 ]  
 [ 0.0021837  -0.0110491  0.41927539]]
```

CoG body [m]: [0.00063905 0.00763669 -0.02125968]

force effectiveness: [1.59449397 1.7918577 1.69308021 1.6933935]

scale-free J/m [m²]:

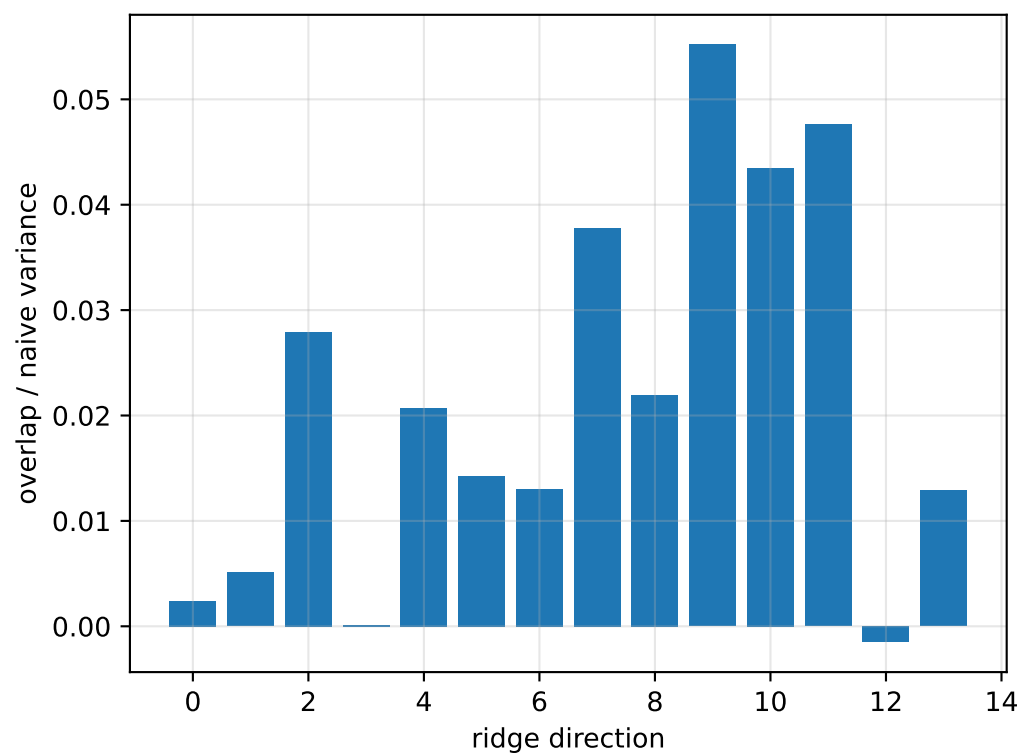
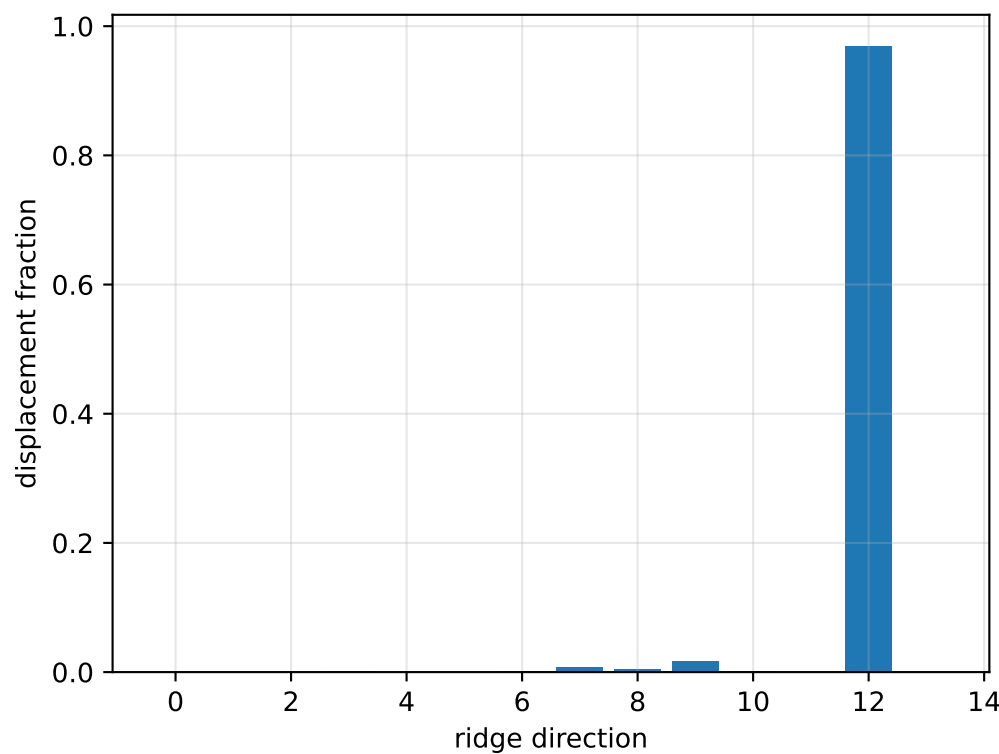
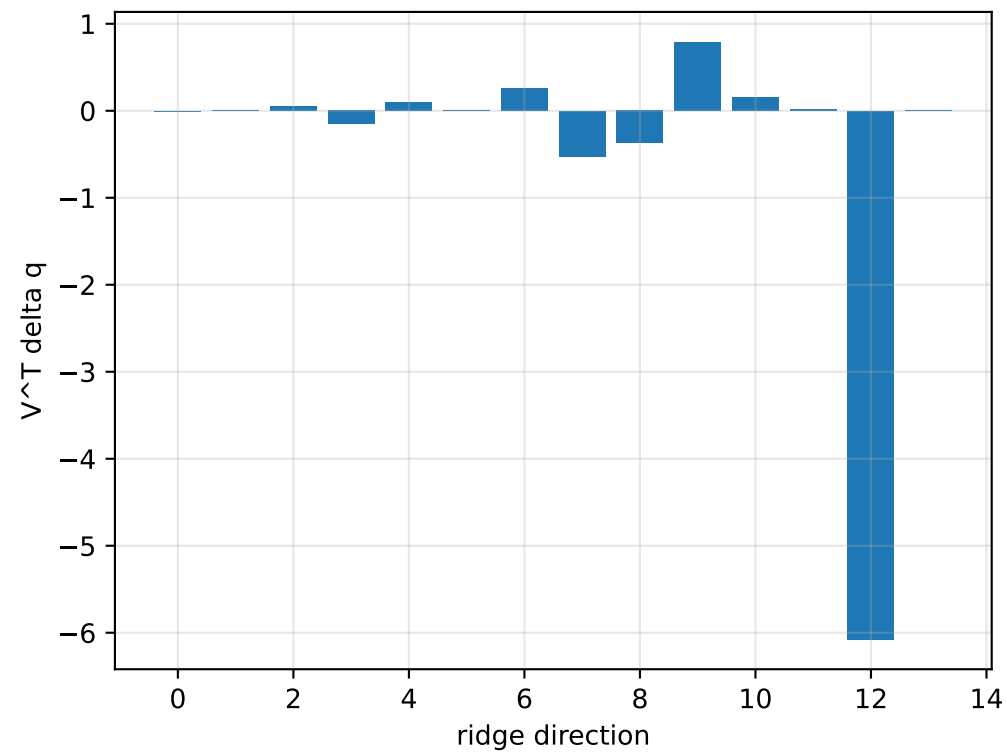
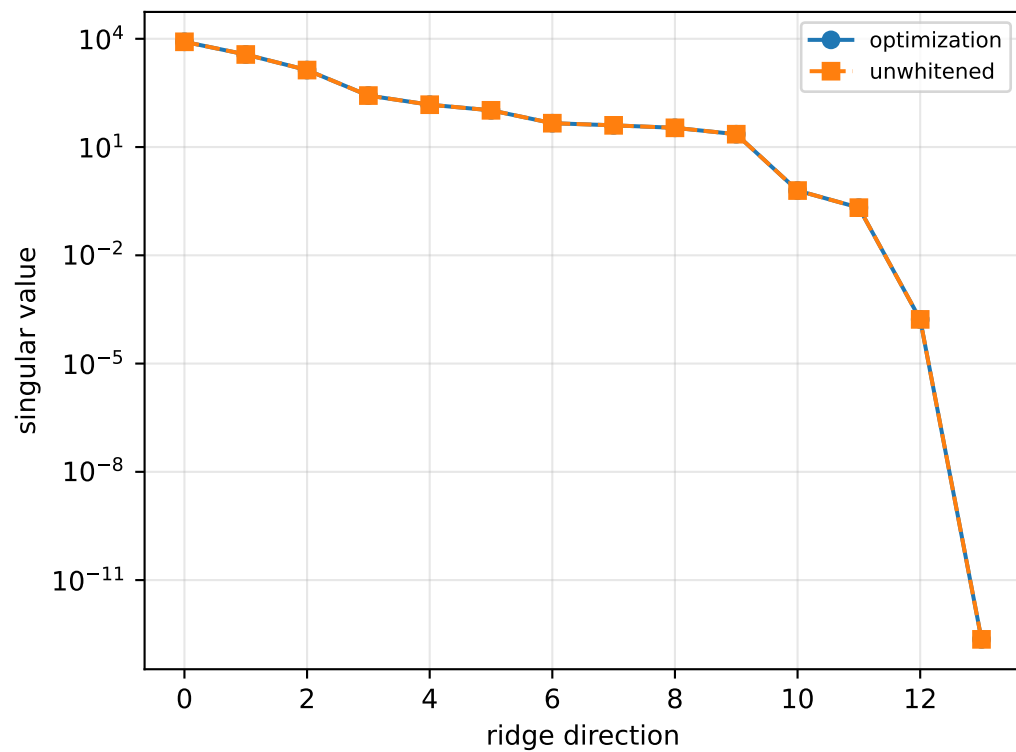
```
[[ 0.02764082 -0.00029193  0.00046506]  
 [-0.00029193  0.06206472 -0.00235312]  
 [ 0.00046506 -0.00235312  0.08929271]]
```

scale-free f/m: [0.33957799 0.38161037 0.36057382 0.36064054]

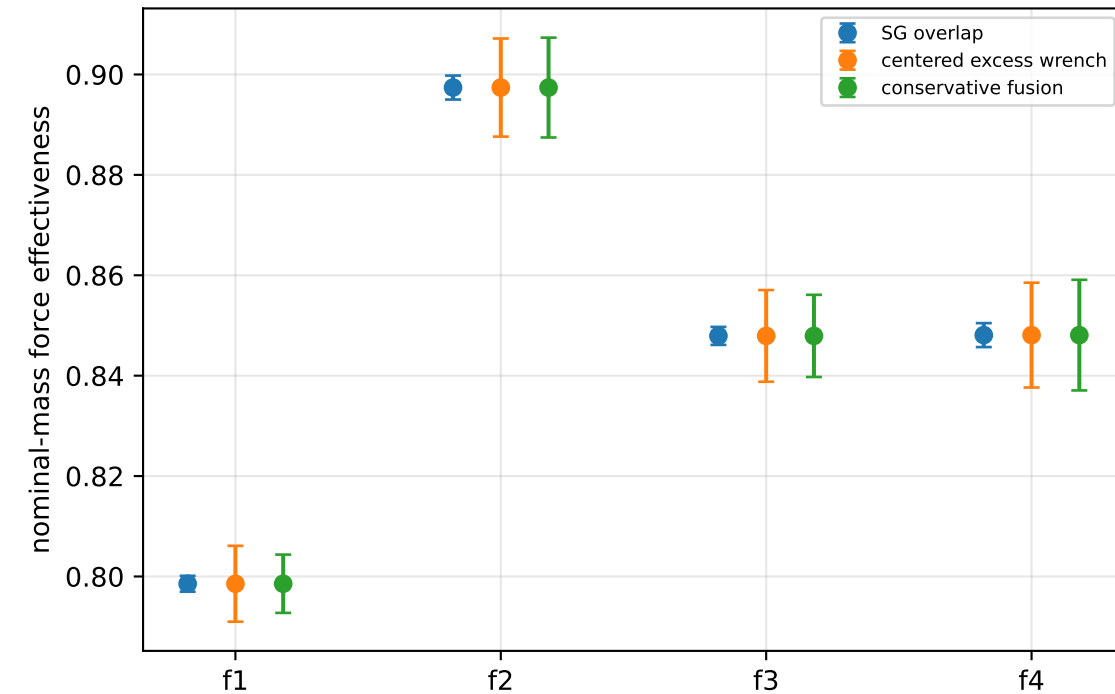
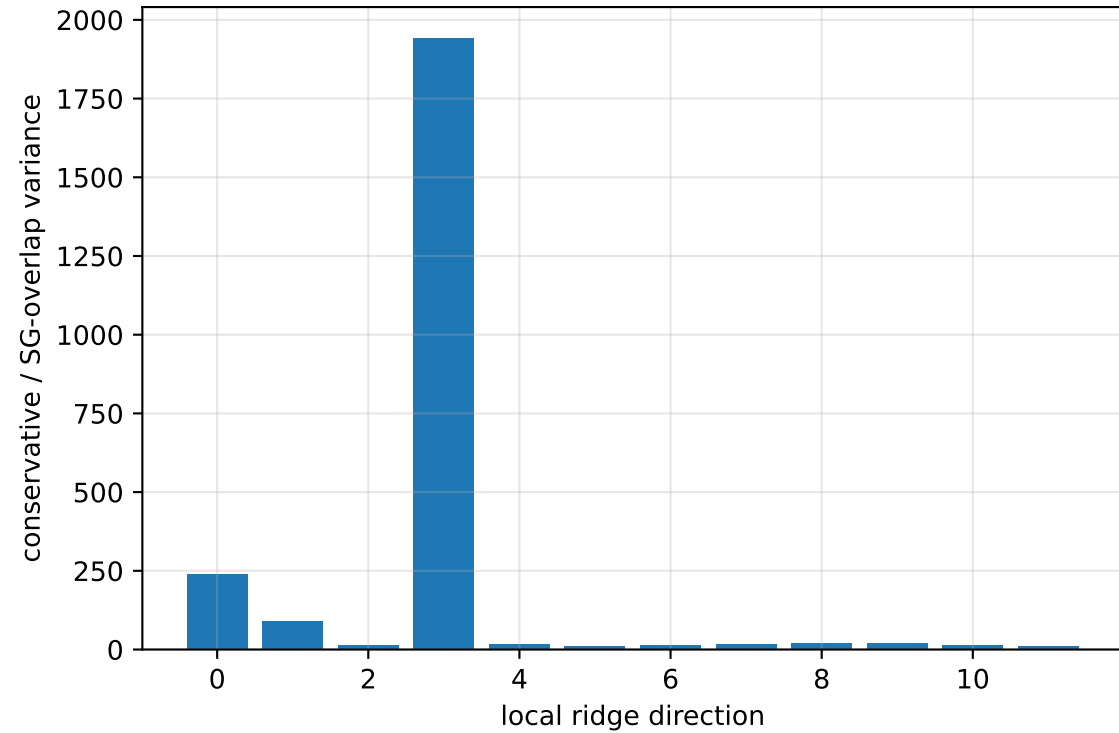
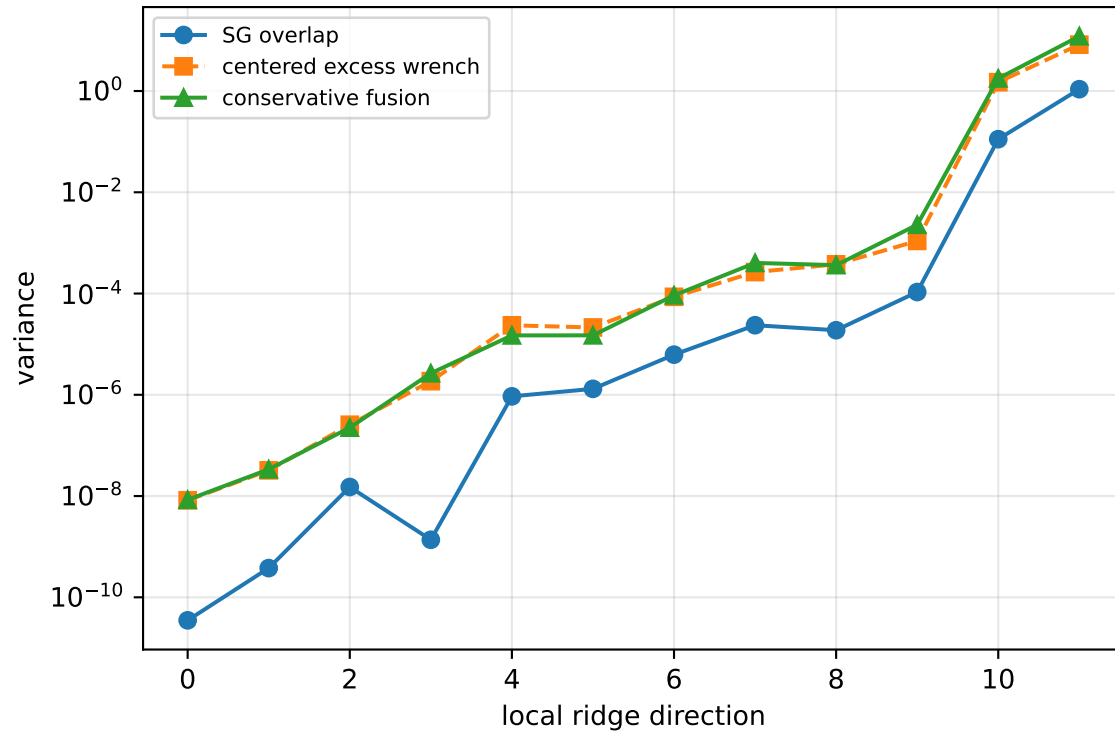
rotor lag cell: (0.196759224, 0.196764469] s

representative: 0.196761847 s

inertia_over_mass_xx_nominal: ridge spectrum (rank 13, nullity 1, || $\mathbf{v}$ ||=2.15e-13)



inertia_over_mass_xx_nominal: Conservative fusion uncertainty



Top three non-gauge ambiguous directions
 $\text{tr}(\mathbf{M}_{\text{res}}) / \text{tr}(\mathbf{M}_{\text{SG}}) = 161.2$
wrench-to-acceleration recovery max error = $3\text{e-}15$
exact scale gauge ridge index = 13 (alignment 1)

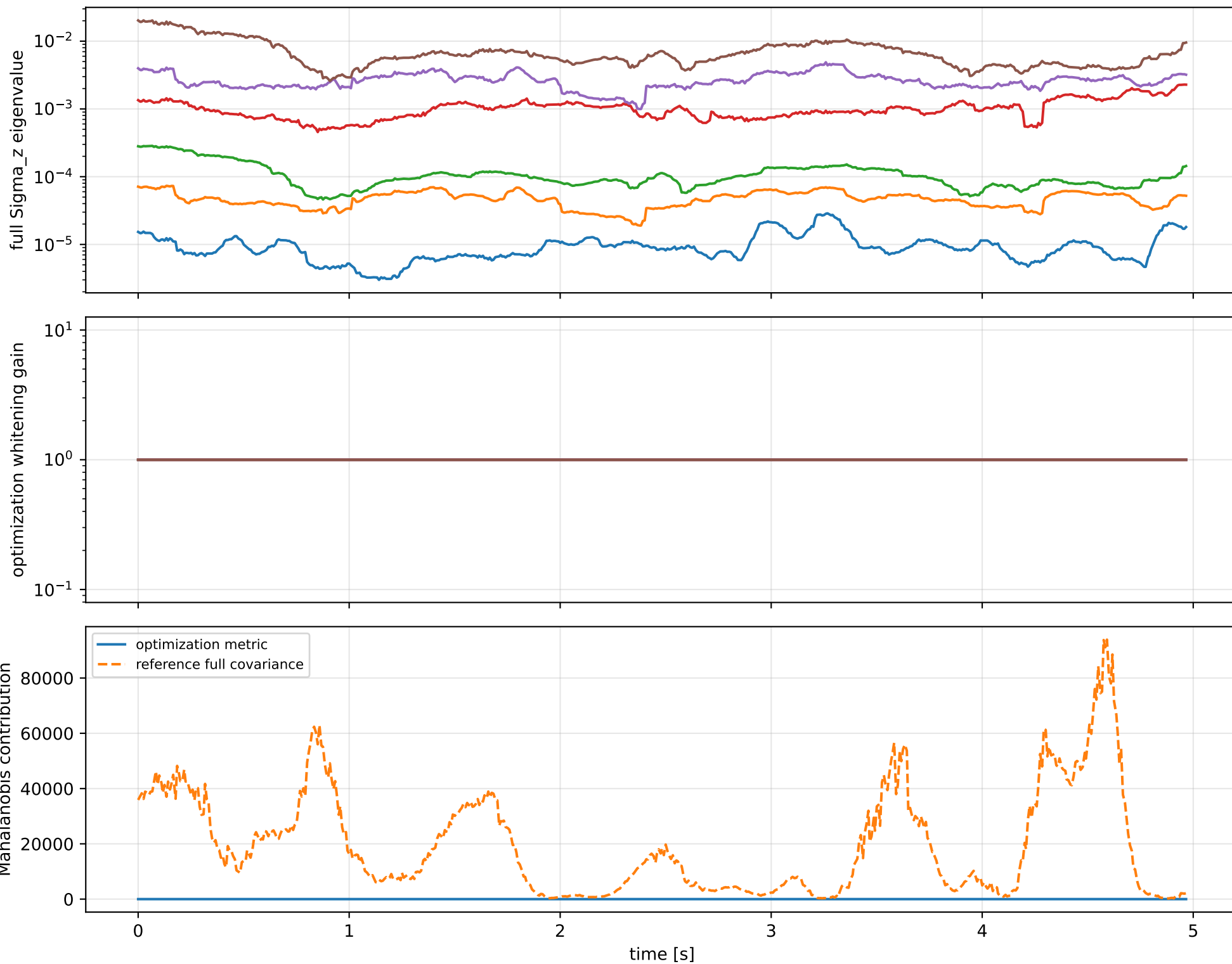
index 11: variance=12.06, ratio=11.1
 $\log_{\text{mass}} = -0.00106$; second_moment_diag_1=-0.0826
second_moment_diag_2=+0.0451; second_moment_diag_3=+0.0428
second_moment_offdiag_12=+0.592; second_moment_offdiag_13=-0.797
second_moment_offdiag_23=-0.0617; cog_x=+9.02e-05
cog_y=-1.57e-05; cog_z=+5.57e-06
log_force_eff_1=-0.001; log_force_eff_2=-0.00103
log_force_eff_3=-0.000295; log_force_eff_4=-0.00189

index 10: variance=1.769, ratio=15.81
 $\log_{\text{mass}} = -0.0254$; second_moment_diag_1=-0.0267
second_moment_diag_2=-0.618; second_moment_diag_3=+0.772
second_moment_offdiag_12=+0.0424; second_moment_offdiag_13=+0.0313
second_moment_offdiag_23=+0.123; cog_x=-0.000525
cog_y=+0.000802; cog_z=+3.1e-05
log_force_eff_1=-0.0267; log_force_eff_2=-0.0305
log_force_eff_3=-0.0249; log_force_eff_4=-0.0197

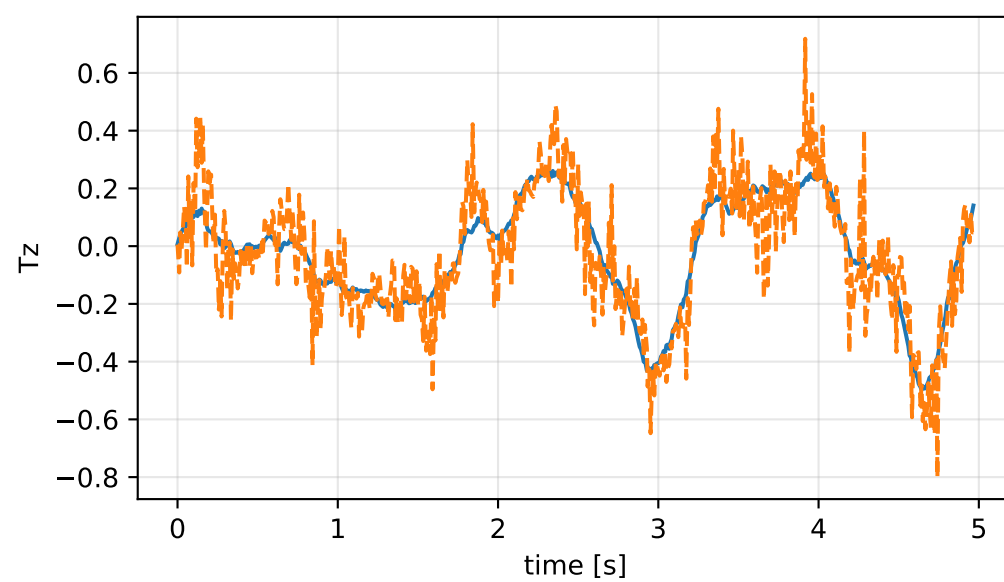
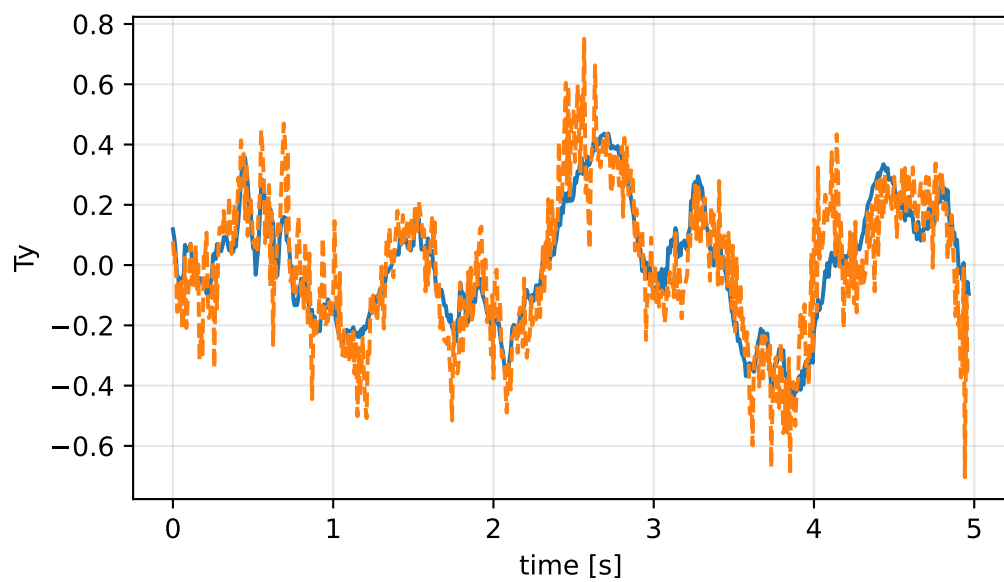
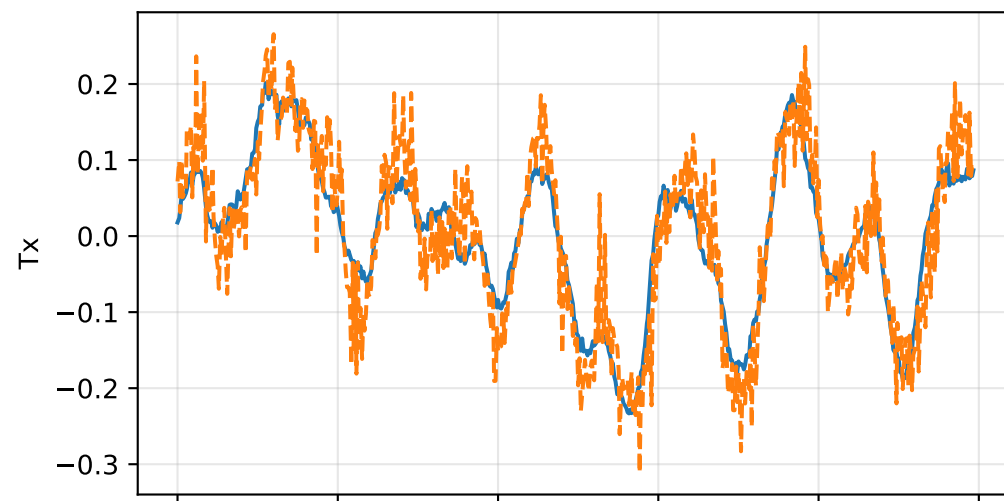
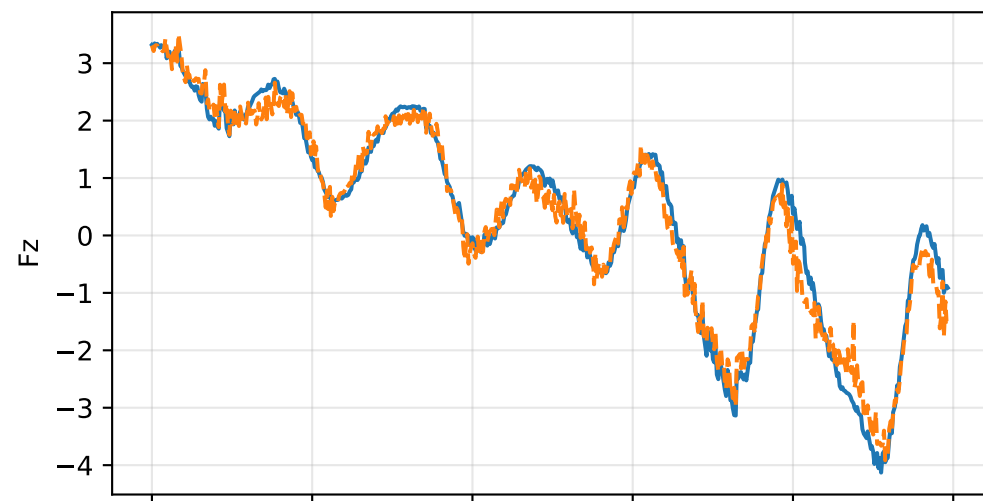
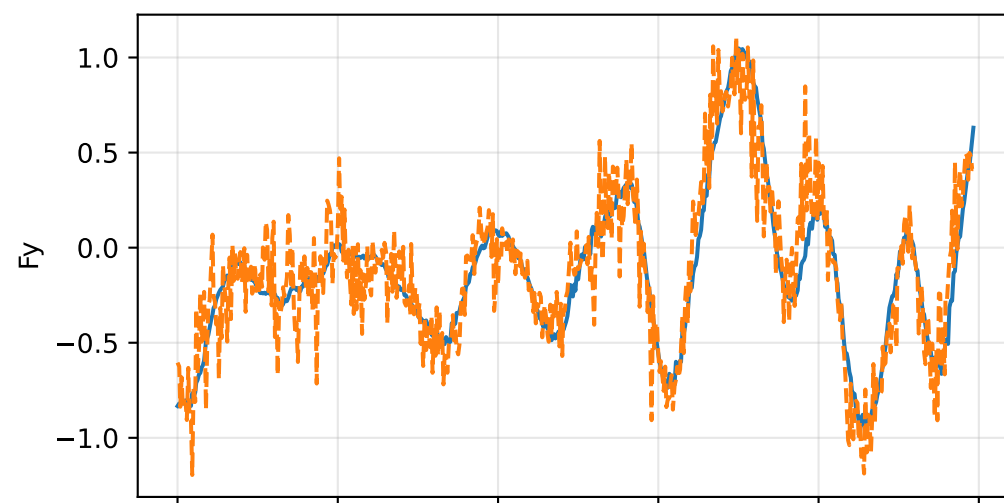
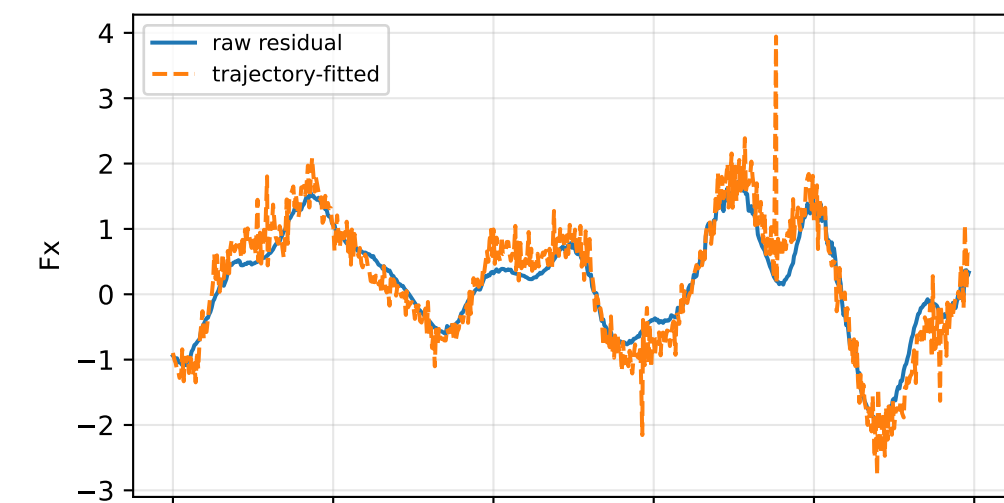
index 9: variance=0.002279, ratio=21.28
 $\log_{\text{mass}} = -0.193$; second_moment_diag_1=+0.107
second_moment_diag_2=+0.543; second_moment_diag_3=+0.306
second_moment_offdiag_12=+0.0919; second_moment_offdiag_13=+0.0597
second_moment_offdiag_23=+0.597; cog_x=+0.0208
cog_y=+0.00203; cog_z=-0.0269
log_force_eff_1=-0.245; log_force_eff_2=-0.14
log_force_eff_3=-0.0442; log_force_eff_4=-0.334

The conservative fusion covariance deliberately retains the existing SG-overlap uncertainty and adds the uncentered empirical residual score second moment without subtracting the SG contribution. It is intended as a conservative fusion distribution, not as a calibrated generative noise covariance.

inertia_over_mass_xx_nominal: optimization vs reference covariance



inertia_over_mass_xx_nominal: wrench / closure (force max 1.47e-14, torque max 8.6e-16)



inertia_over_mass_xx_nominal: optional parameter-prior audit

Optional physical parameter prior

The parameter prior is optional external information. The default estimator is prior-free.

name: pseudo_condition_inertia_over_mass_xx_nominal

role: pseudo_conditioning_ablation

source: /home/leus/catkin_ws/src/ProbTF-demo/ros/examples/grape-param-estim/minimal/config/priors/pseudo_conditioning/scalar/inertia_over_ma

SHA256: d50f14634ccbffa93995c113d524958bbedaa410a60c8ea864b4c5cd041abdb4

data objective: 416.520626131

prior objective: 7.83301150662e-05

total objective: 416.520704461

This configuration uses an intentionally tight artificial standard deviation for an ablation-style pseudo-conditioning experiment; it is not a calibrated physical uncertainty.

factor: inertia_over_mass_xx_nominal (inertia_over_mass_m2)

components: xx

target: [0.02764081]

std: [1.e-06]

final: [0.02764082]

error: [1.25163984e-08]

standardized residual: [0.0125164]

factor objective: 7.83301150662e-05